Yarmouk University

Hijjawi Faculty for Engineering Technology Department of Electrical Power Engineering

Load frequency control (LFC)

Done by:

Obadah Mohammed Abudoleh

Supervisor:

Dr. Assmaa alhatimi

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ABSTRACT

In an interconnected power system, even a small, sudden change in load in one area can lead to frequency fluctuations across the entire network and affect the power flow through the tie lines. The primary goals of Load Frequency Control (LFC) are to keep the system's frequency stable and maintain the desired power output (in megawatts), while also managing power exchange between different control areas. To achieve this, an LFC setup uses a well-designed control system that helps bring the frequency and tie-line power back to their normal or near-normal levels after any load disturbance.

Traditional controllers are commonly used for this purpose, but they come with certain limitations. They tend to be slow to respond, don't account for the nonlinear behavior of many power system components, and can be tricky to tune especially when adjusting the integrator gain for varying operating conditions.

On the other hand, advanced control systems offer clear benefits. They respond faster and ensure better system stability. This study explores the use of advanced control strategies like optimal controllers, optimal compensators, and IMC-PID control in managing the LFC of a two-area power system. These optimal controllers and compensators are particularly useful because they work well even without complete state feedback and can handle both processes and measure noise. The IMC-PID controller delivers enhanced performance and deals effectively with system nonlinearities.

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Chapter 1: Introduction

Load Frequency Control (LFC) is considered one of the most crucial aspects of today's power systems. In simple terms, it's a method used to keep the electrical grid stable and reliable by regulating the frequency of the power supply. Its main role is to help maintain the balance between power generation and consumption, ensuring the frequency stays within acceptable limits.

Whenever there's a mismatch between how much power is being generated and how much is being used, the system's frequency can shift slightly. These deviations, even if small, can lead to equipment malfunctions, energy loss, or in extreme cases, a complete system failure.

LFC systems use advanced control techniques to address this imbalance in real-time. They adjust generator output and manage power flow between interconnected regions to keep everything in check. As a result, the power supply remains consistent and secure, helping maintain both the quality and performance of electricity delivered to consumers.

1.1 Load Frequency Control Problem

A power system consists of several control areas that are linked together through tie lines. The generators within each area work in coordination, adjusting their speeds collectively to keep the overall system frequency and power angles stable even when conditions change. However, if there's a sudden shift in load in one area, it can lead to fluctuations in both frequency and the power flowing through the tie lines.

Main Objectives of Load Frequency Control (LFC):

1. Keep the system's actual frequency steady and ensure the desired power output (measured in megawatts) is maintained.

2. Manage power flow between different control areas by regulating changes through the tie lines.

In a single-area power system, even a small change in load can disrupt the power balance. At first, this imbalance is managed by drawing on the system's stored kinetic energy, which causes a drop in frequency. A new equilibrium is eventually reached when the higher load ends up using less power due to the lower frequency, but this comes at the expense of that frequency drop.

In cases of larger imbalances, the governor steps in by increasing the generator's output to help restore balance. This reduces the reliance on kinetic energy, although it doesn't eliminate the frequency deviation.

This process of adjusting generator output to respond to load changes is known as primary control, and it typically kicks in within 10 to 12 seconds after a disturbance. However, even after the governor has done its part, the system frequency might still not return to its original setpoint.

To fine-tune the frequency and bring it back to its exact predefined value, secondary control is used. This is usually handled by integral controllers, and while it's slower than primary control, it plays a key role in restoring long-term stability.

In a two-area interconnected power system, any load change in one area doesn't just affect that area alone, it impacts both. This means the control system needs data from all areas to bring the frequencies and tie-line power flows back to their target values.

1.2 Interconnected Power Systems

From a practical standpoint, Load Frequency Control (LFC) is even more crucial in interconnected power systems than in isolated (single-area) ones. However, having a solid understanding of the theory and principles

behind isolated systems is still important, as it forms the foundation for comprehending how interconnected systems operate.

Today, nearly all major power systems are linked with neighboring regions, making LFC a shared and more complex challenge across multiple areas. The core operating principles of an interconnected power system are outlined below:

1. Under normal conditions, each control area is responsible for meeting its own power demands, except for the specific portion of load that's been scheduled to be shared with other areas, based on mutual agreements.
2. All areas must agree on using control strategies and equipment that remain effective not only during normal operations but also in the event of disturbances or abnormal conditions within the power system.

1.3 Advantages of Interconnection

Effect of size: This is one of the key benefits of having an interconnected power system. When a sudden load is added, the initial energy needed to meet the demand is drawn from the system's kinetic energy. In larger systems, there's typically more stored kinetic energy available, which means the frequency drop is smaller and less noticeable. On the other hand, in a single-area (isolated) system, the same load change would cause a much larger frequency drop due to the limited available energy.

1.4 Need for reduced reserve capacity

Peak demand can occur unpredictably at different times across various regions. In large power systems, the difference between peak load and average load tends to be smaller than in smaller, isolated systems. As a result, interconnected power systems can reduce the need for extra reserve capacity

by coordinating scheduled energy exchanges between areas.

When two control areas are linked by a tie line, the system is known as a two-area interconnected power system. As shown in Figure 1.1, each area is primarily responsible for supplying power to its own region, while the tie line allows for power exchange between them, enhancing overall system reliability and efficiency.

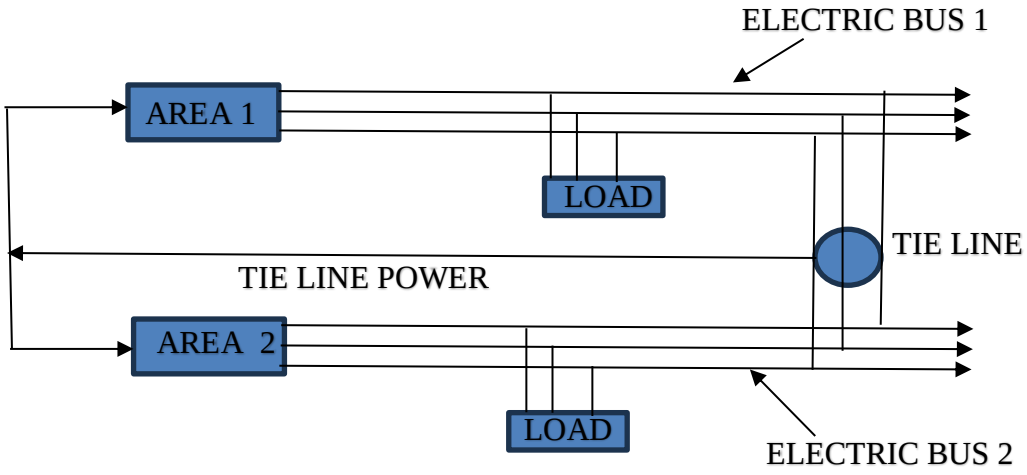


Figure 1.1Two area interconnected power system

In a two-area power system, it is typically assumed that each individual area is strong, while the tie line connecting them is relatively weak. This means that each area maintains a uniform frequency, reflecting the idea that the internal network of each area is "strong" or "stiff." An interconnected power system can consist of any number of control areas, not just two.

1.5 Major Drawbacks of Conventional Integral Controller:

The drawbacks of conventional integral controllers can be summarized as follows:

1. Slow Response: These controllers tend to react slowly, which limits their effectiveness in dynamic power system environments.

2. Inability to Handle Nonlinearities: They do not account for the inherent nonlinear behavior of various power system components. Examples include governor dead-band effects, generation rate constraints (GRCs), and the characteristics of reheat-type turbines commonly used in thermal power plants.
3. As power system loads shift throughout the day, the system's operating point also changes the natural and well-known behavior in power networks. To keep performance at its best, the integrator gain should ideally be adjusted to match these ongoing changes. The challenge lies in finding the right balance: a gain that ensures quick recovery from disturbances without causing excessive overshoot. In practice, maintaining this balance is quite difficult. That's why integral controllers are generally fixed, they're tuned to working well at one specific operating point, but their effectiveness drops when conditions shift. Because of this, the control strategy used must reflect the dynamic nature of the power system. Advanced controllers are much better equipped to handle these changes, offering more adaptive and reliable control.

1.6 Need of Advance Control Technique

The use of advanced control techniques plays a vital role in improving Load Frequency Control (LFC) in modern power systems. As today's power networks grow more complex and operate in increasingly uncertain and less structured environments, there is a clear need for smarter, more efficient control methods to ensure stable, secure, and cost-effective operation.

What sets advanced control strategies apart is their adaptability to changing conditions and their ability to respond quickly to system disturbances. Some of the well-known techniques in this area include Optimal Control, Pole Placement, Linear Quadratic Regulator (LQR), Linear Quadratic Gaussian (LQG), Robust Control, Sliding Mode Control, and

Internal Model Control (IMC). In this study, LQR, LQG, and IMC have been applied to enhance the LFC performance of the power system.

1.7 Modeling of Power System For LFC:

Accurate modeling of power systems is essential for effective Load Frequency Control (LFC) analysis. In this study, a two-area power system model has been used, specifically representing a thermal-thermal system without reheaters. This type of model provides a solid foundation for evaluating control strategies. The structure of the system is illustrated in Figure.

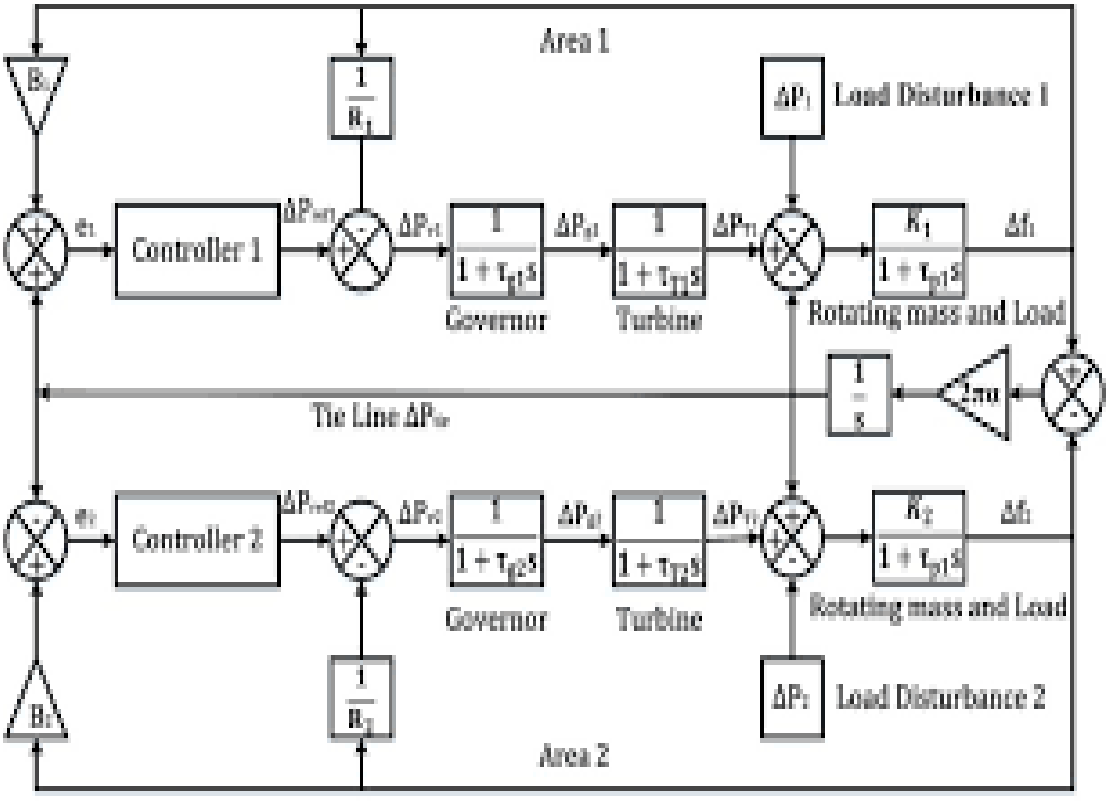


Figure 1.2:Two area thermal (non reheat) power system with integral controller.

The model discussed here represents the integral control scheme for an interconnected power system. It uses state-space modeling of the specified system to facilitate the design of optimal controllers and analyze system stability.

Figure 1.2 shows the block diagram of a two-area thermal non-reheat

power system equipped with integral controllers. The system's state equations are derived from the transfer functions of blocks labeled 1 through 7.

This model depicts a two-area power system where each area is connected by a tie line, which has its own dynamic behavior. Both control areas are identical, each containing thermal non-reheat turbines.

Within each control area, there are three main blocks: a governor block, a turbine block, and a power system block (representing the load). Each of these blocks includes an integral controller, making a total of nine blocks in the entire system.

Each control area is composed of three blocks: a governor block, a turbine block, and a power system block (which represents the load). Each block includes an integral controller. Therefore, the entire system consists of a total of nine blocks.

1.8 Objectives

The two primary objectives of Load Frequency Control (LFC) are:

1. To maintain the actual system frequency and the desired power output (in megawatts) within the interconnected power system.
2. To regulate the power flow changes through the tie lines between different control areas.

Chapter 2: Load Frequency Control Stages

Load Frequency Control (LFC) plays a crucial role in regulating frequency fluctuations and ensuring zero steady-state error. According to the IEEE 446-1995 standards, consumer equipment can typically tolerate frequency deviations of up to ±0.5 Hz which translates to about ±1% for a 50 Hz system and ±0.833% for a 60 Hz system. LFC is generally divided into three stages: primary, secondary, and tertiary control mechanisms.

2.1 Primary Control Mechanism

As illustrated in Figure 2.1, steam flows into the turbine, which then supplies fuel to the generator, causing it to produce electricity. The system uses Automatic Generation Control (AGC), which includes feedback loops for both the Automatic Voltage Regulator (AVR) and Load Frequency Control (LFC). When the frequency sensor detects any deviation between the turbine's speed and the generator's output frequency, it signals the LFC system to take corrective action.

The valve control adjusts the amount of steam fed to the turbine based on the LFC's output, helping to restore the frequency to its desired level. This frequency control maintains a balance between power generation and consumption, ensuring smooth and reliable parallel operation of the power station.

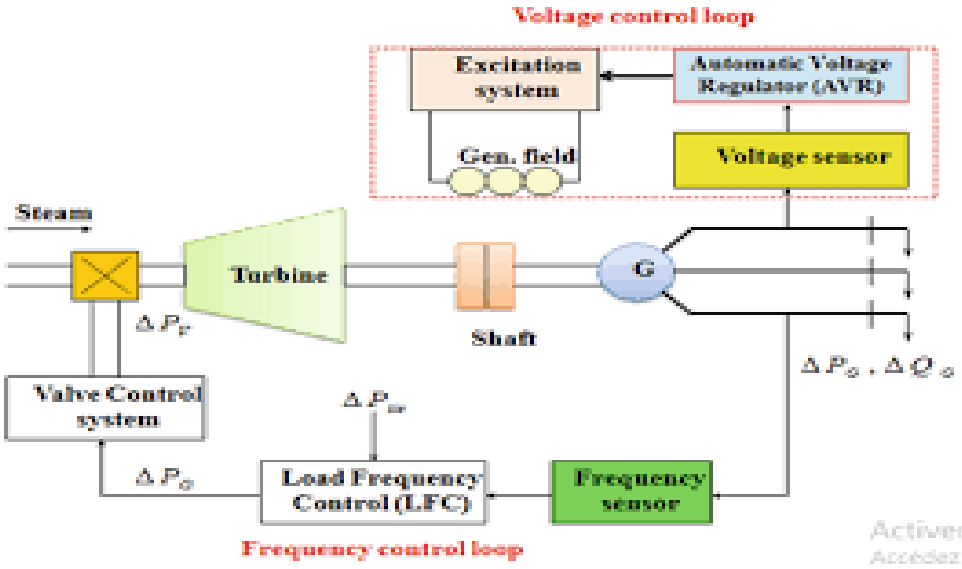


Figure 2.1: Automatic Generation Control for a generator

2.1.1 Speed Governing System

The frequency regulation achieved through the governors' action is known as primary control. This process is automatic: when the system experiences a speed drop due to an increased load, the governor responds by increasing the input such as admitting more steam into the turbine in steam power plants which raises the output power.

In the system described, the governor continuously monitors the mechanical speed and adjusts the steam input valve accordingly. This adjustment changes the mechanical power output to offset load variations, helping to bring the frequency back to its minimum acceptable level. As shown in Figure 2.2, the speed governor is located between the motor and a hydraulic amplifier, which acts like an electrical switch in this setup.

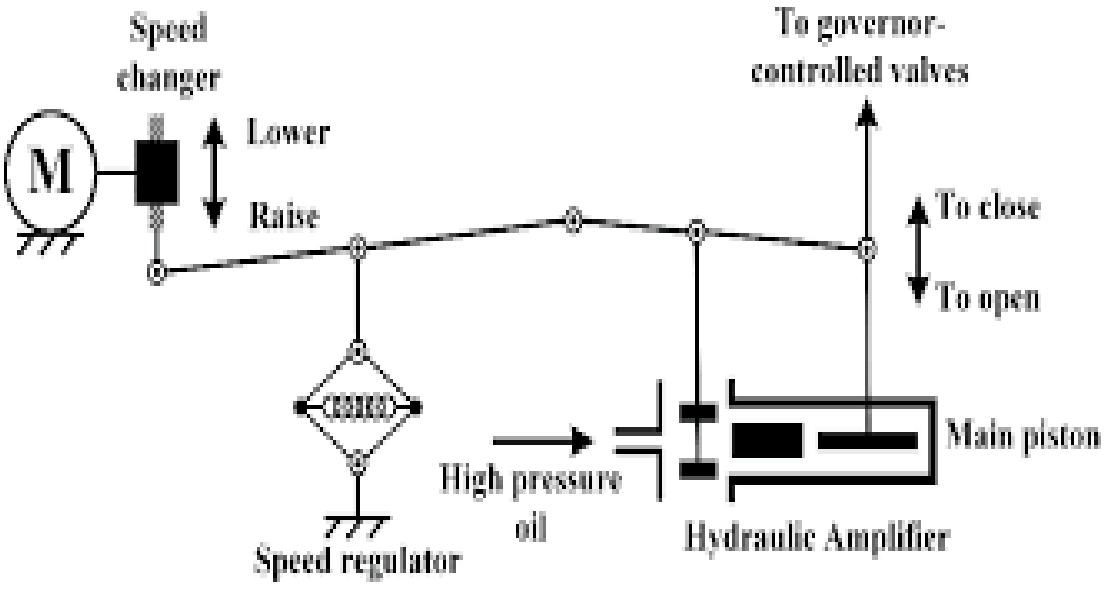


Figure 2.2: Speed governing system.

2.1.2 Prime Mover

In the primary stage of Load Frequency Control (LFC), the prime mover is the initial source of energy that drives power generation. This could be a steam turbine or any other type of fuel-based engine. The prime mover converts mechanical energy into electrical energy, which is then supplied to the power grid.

The speed of synchronous and induction motors commonly used as prime movers in industry is directly tied to the supply frequency. For synchronous motors, the speed is calculated as 120 f/P, where f is the supply frequency and P is the number of poles. Because of this relationship, any change in supply frequency causes the motor speed to vary. This variation is undesirable, especially for industries that rely on constant-speed drives for their processes.

2.2 Secondary LFC Mechanism

When there's a sudden loss or change in load, the governor acts to control the generator's speed. However, relying on governors alone for

frequency control isn't enough, which is why secondary control is needed. In this stage, the central load dispatcher adjusts the loading on different power plants to help stabilize the system frequency more effectively.

Secondary control is crucial for the smooth operation of multi-area power systems. It helps maintain the balance between generation and load within each control area, manages the power exchanged between areas, and greatly reduces the effects of unpredictable changes caused by the variability in power generation. This not only improves overall stability but also boosts the reliability of the power system. As power systems continue to evolve especially with the growing share of renewable energy sources secondary control becomes even more important.

2.2.1 Limitations on Isolated Area.

Isolated microgrids generally have fewer regulation capabilities compared to large, interconnected power systems, which makes maintaining their stable operation much more challenging.

Because of constant load fluctuations and the variable nature of renewable energy sources (RES), isolated microgrids often face frequency deviations.

Since these microgrids rely heavily on renewables like solar and wind, their stability can be affected. While PID controllers are commonly used to manage power input variations and keep the grid stable, they can struggle when sudden changes or disturbances occur as we discussed earlier in Chapter 1.

2.2.2 Multi-Area Interconnected in LFC.

Multi-area power systems are extensive networks made up of interconnected grids, each acting as its own control area with local generation and load demands. To keep these systems stable and reliable, specific control

methods and regulations are essential.

Secondary control, often called automatic generation control (AGC), is especially important in these multi-area systems. Its main job is to adjust generator outputs within each control area to balance the load while making sure the power exchanged between areas follows the agreed schedules. Unlike primary control, which acts quickly to stabilize frequency deviations over a short period, secondary control takes place over a longer timeframe. It corrects any steady state errors and helps maintain the overall system's stability and integrity. Figure 2.3 illustrates the timing differences between these two control stages.

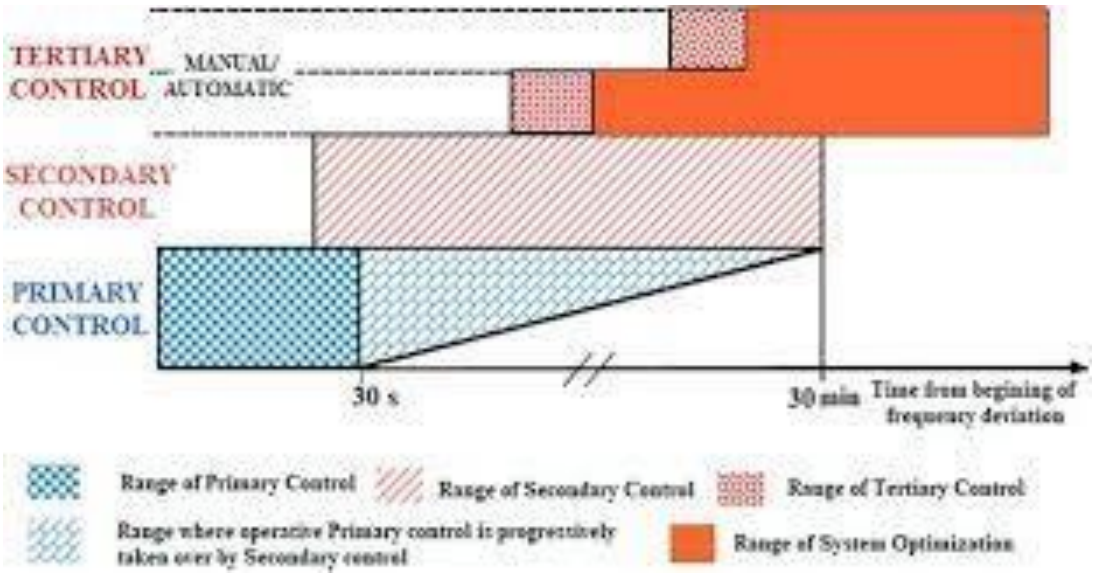


Figure 2.3: The frequency deviation control levels

So, what are the advantages of multi-area interconnected area in load frequency control? Secondary control is a crucial part of running multi-area power systems. It helps keep the balance between power generation and demand within each control area, manages the flow of power between different areas, and greatly minimizes the effects of unpredictable changes in power generation. This all contributes to making the power system more stable and reliable.

2.2.3 Adjusting of Nominal Frequency:

Frequency stability starts with the primary control, which acts fast to keep the system stable but might settle the frequency at a value slightly different from the standard like around 50 Hz or 60 Hz. That's where secondary control steps in, working to bring the frequency precisely back to its nominal value. This adjustment is essential to ensure that all electrical devices operate correctly, and the power grid remains stable.

2.2.4 Inter-area Power Balance:

Load Sharing: In interconnected multi-area power systems, secondary control ensures that each area takes on its proper share of the total system load and maintains the scheduled power exchanges between areas.

2.2.5 Area Control Error (ACE):

Controlling power generation in each area to minimize the Area Control Error (ACE) is a primary function of secondary control. This not only ensures that power flows between areas remain within the agreed system limits (known as the interchange schedule) but also helps prevent imbalances, the root cause of frequency deviations. Such imbalances can lead to severe consequences, including large-scale emergency shutdowns of generating units and widespread grid failures.

2.2.6 Economic Operation of Power Systems:

Optimal Generation Dispatch: Effective secondary control ensures that the most cost- efficient generation units are utilized to meet the load demand, thereby minimizing overall system operating costs.

Renewable Energy Uncertainty: Unfortunately, multi-area power systems face significant challenges due to the increasing integration of

renewable energy sources (RES). Solar and wind generation are inherently variable and fluctuate with changing environmental conditions, introducing uncertainty into the system. As a result, reliance on isolated single-area power systems remains important, as they can provide more predictable and stable operation when managing these uncertainties.

2.3 Tie Lines

Tie lines play a crucial role in connecting two balanced authority areas or control areas, as they facilitate power sharing between interconnected areas or control areas. Tie lines may be represented by actual physical transmission lines, series devices, or transformers. Tie lines are classified into two main types: external and internal tie lines.

2.3.1 External Tie Lines

External tie lines are used to connect two or more areas within the power system. They are commonly employed to exchange electricity between countries or different local power generating companies, allowing the transfer of electrical power from locations with surplus generation to those with higher demand. This optimizes load distribution, helps prevent outages, enhances the stability of the internal network, and improves overall operational efficiency.

2.3.2 Internal Tie Lines

Internal tie lines, unlike external ones, connect generation stations or distribution stations within the same local network. They are used to distribute electricity between stations in the same network, enhancing load distribution and preventing outages. Internal tie lines contribute to improving the stability of the electricity network and increase efficiency by balancing loads and facilitating power exchange within the area.

Chapter 3: MODELLIN OF LFC

Modelling of Load Frequency Control

In modelling load frequency control, the focus is on each part of the block diagram of a steam power area. This includes detailed modelling of:

- **Single area systems**, where only one control area is considered.
- **Two interconnected areas**, which represent multi-area power systems connected by tie lines.

Each block in the system represents key components such as the **governor**, **generator**, **load**, and **prime mover**. Understanding the dynamics of these individual components is essential for designing effective Load Frequency Control strategies.

3.1 Generator Model

Under normal operating conditions, the relative position of the rotor axis and the resultant magnetic field axis is fixed. The angle between the two is known as the power angle or torque angle.

During any disturbance, the rotor speed will accelerate or decrease with respect to the synchronously rotating air gap, and relative motion begins. The equations used to describe this motion are called swing equations.

Consider a synchronous generator developing an electromagnetic torque T_e and running at synchronous speed ω_{sm}. If Tm is the driving mechanical torque under steady-state operation and losses are neglected, then the torque balance equation is:

$$\mathbf{Tm=Te} \qquad \text{(Eq.1)}$$

$$\mathbf{Ta = Tm-Te} \qquad \text{(Eq.2)}$$

And if ***J*** is the combined moment of inertia of the prime mover and generator, neglecting frictional and damping torques, from the laws of rotation we have:

$$J\ \frac{d^2\delta_m}{dt^2} = T_m - T_e \qquad \text{(Eq.3)}$$

Here, δ_m is the mechanical angular displacement of the rotor relative to a reference axis on the stator, and δ is the corresponding electrical power angle, which is more convenient to use in power system analysis.

$$\delta = \frac{p}{2}\delta_m \qquad \text{(Eq.4)}$$

$$\omega = \frac{p}{2}\ \omega_m \qquad \text{(Eq.5)}$$

Now define the important quantity known as *H* constant or the per unit inertia constant:

$$H = \frac{kinetic\ energy\ in\ MJ\ at\ rated\ speed}{machine\ rating\ in\ MVA} = \frac{W_k}{S_B} \qquad \text{(Eq.6)}$$

Applying the swing equations of synchronous machine to a small distribution, we have:

$$\frac{2H}{\omega_s}\frac{d^2\Delta\delta}{dt^2} = \Delta P_m - \Delta P_e \qquad \text{(Eq.7)}$$

Without speed expressed in per unit and taking Laplace transform we obtain:

$$\Delta\Omega(s) = \frac{1}{2H_s}[\Delta P_m(s) - \Delta P_e(s)] \qquad \text{(Eq.8)}$$

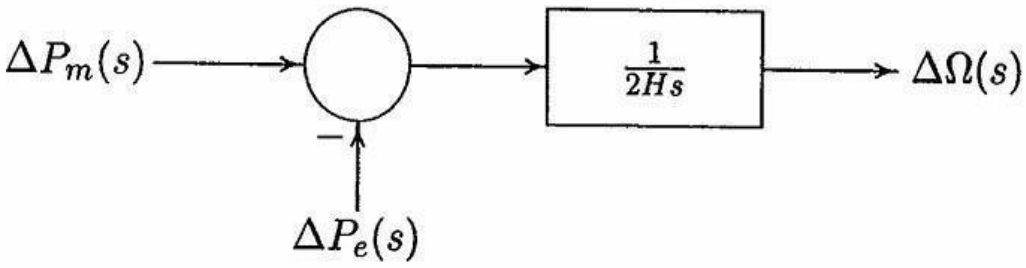


Figure 3.1: Generator Block Diagram

3.2 Load Model

The speed-load characteristics of a composite load are approximated by considering how electrical power varies with frequency. Resistive loads like lighting and heating have power independent of frequency, while motor loads are sensitive to frequency changes. The degree of sensitivity depends on the combined speed-load behavior of all connected devices.

$$\Delta P_e = \Delta P_L + D\Delta\omega \qquad \text{(Eq.9)}$$

ΔP_L represents the load change that is not sensitive to frequency, while D·Δω represents the load change that is frequency sensitive. Here, D is defined as the percentage change in load divided by the percentage change in frequency.

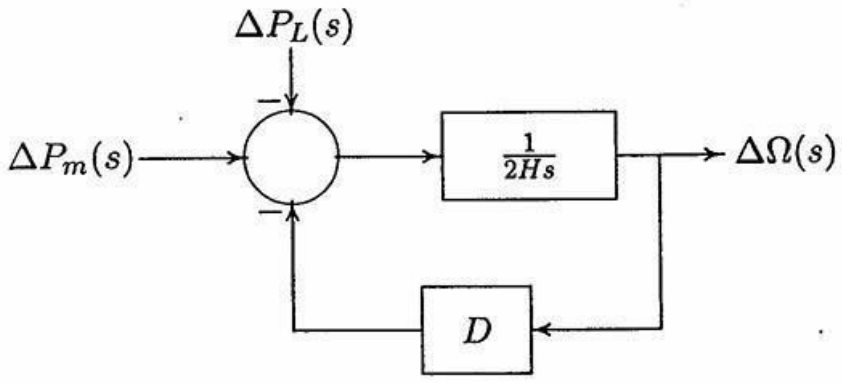


Figure 3.2: Generator and Load Block Diagram

3.3 Prime Mover Model

The source of mechanical power, commonly referred to as the prime mover, may include hydraulic turbines at waterfalls, steam turbines powered by coal, gas, or nuclear fuel, and gas turbines. The simplest model for a non-reheat steam turbine prime mover can be represented by a single time constant τ_T (ranging from 0.2 to 2.0 seconds), leading to the following transfer function:

$$G_T(s) = \frac{\Delta P_m(s)}{\Delta P_V(s)} = \frac{1}{1+\tau_T^2} \qquad \text{(Eq.10)}$$

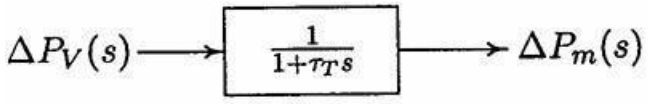


Figure 3.3: Simple Block Diagram Of Non-Reheat Steam Turbi

3.4 Governor Model

When the electrical load on a generator suddenly increases, the electrical power demand surpasses the mechanical power input. This shortfall is compensated for by the kinetic energy stored in the rotating system. As kinetic energy decreases, the turbine speeds and, consequently, the generator frequency drops. This speed change is detected by the turbine governor, which adjusts the turbine input valve to modify the mechanical power output, stabilizing the speed at a new steady state. Early governors, such as Watt governors, sensed speed through rotating flyballs and provided mechanical responses to speed changes. In contrast, most modern governors use electronic sensors to detect speed variations. This model consists of four main parts:

3.4.1 Speed Governor

The essential components include centrifugal flyballs that are driven directly or via gearing by the turbine shaft. This mechanism produces upward and downward vertical movements proportional to the changes in speed. Figure 3.5 illustrates the block diagram of the speed governor.

3.4.2 Linkage Mechanism

These are the linkages that transfer the flyballs' movement to the turbine valve via a hydraulic amplifier and also provide feedback from the turbine valve's movement

3.4.3 Hydraulic Amplifier

Because operating the steam valve requires exceptionally large mechanical forces, the governor's movements are converted into high-power forces through multiple stages of hydraulic amplifiers.

3.4.4 Speed Changer

The speed changer includes a servomotor that can be controlled either manually or automatically to schedule the load at nominal frequency.

In order to ensure stable operation, governors are designed to allow the speed to decrease as the load increases. The steady-state behavior of this type of governor is illustrated in Figure 3.4.

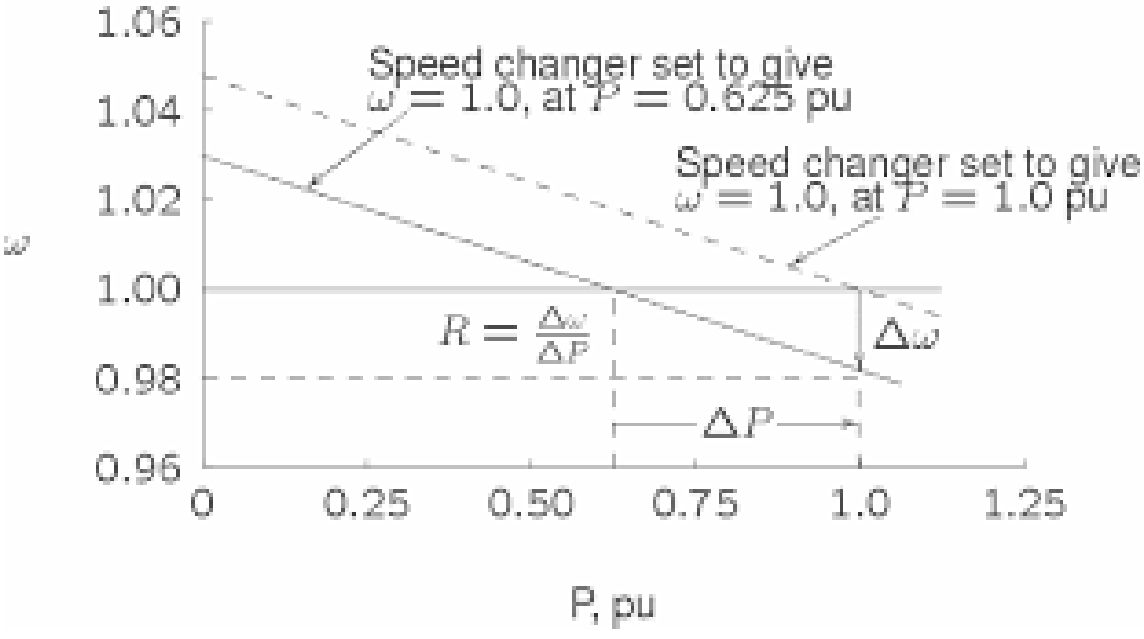


Figure 3.4: Governor Steady-State Speed Characteristics

The slope of the curve represents the speed regulation, denoted as **R**. Typically, governors have a speed regulation of about 5-6% from no load to full load. The speed governor functions as a comparator, where its output, ΔPo, is the difference between the reference set power, ΔPres, and the power Δω derived from the governor's speed characteristics.

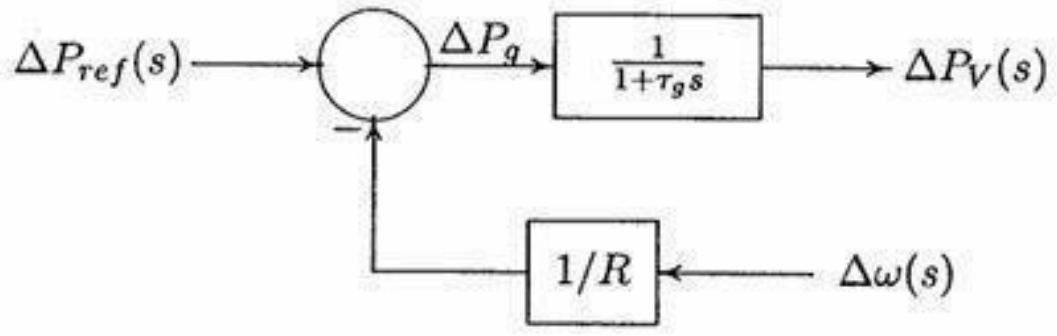


Figure 3.5: Governor Speed Block Diagram

3.5 Automatic Generation Control

When the load on a power system rises, the turbine speed initially decreases before the governor can adjust the steam input to meet the new demand. As the speed deviation lessens, the error signal decreases, and the governor flyballs move closer to the position required to maintain a constant speed, though a steady-state offset from the set point remains. To bring the speed or frequency back to its nominal value, an integrator is added to track the average error over time—this is called reset action.

Automatic Generation Control (AGC) automatically adjusts generation to maintain nominal frequency as system load varies. In interconnected systems with multiple control areas, AGC allocates loads among systems, stations, and generators to maximize economic operation and control tie-line power flows while keeping frequency uniform. During large disturbances, AGC is bypassed in favor of emergency controls.

In a single-area system, a load change causes a steady-state frequency

deviation due to governor speed regulation. To eliminate this frequency deviation, an integral controller acts on the load reference setting by adjusting the speed set point. This increases the system type by one, driving the steady-state frequency deviation to zero. The system with this secondary loop is illustrated in Figure 3.6, and the integral controller gain must be tuned to achieve a satisfactory transient response.

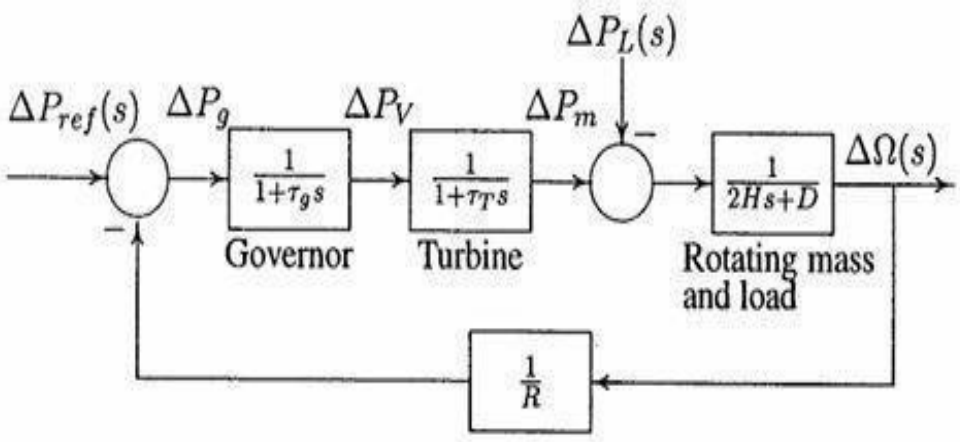


Figure 3.6: AGC For an Isolated Power System

Chapter 4: SIMULINK MODEL FOR THREE-AREA LFC

4.1 Simulink Model for Three-Area Load Frequency Control (LFC)

The system parameters for the three-area LFC model are provided in the table below. The base power for the system is 1000 MVA.

i – j	Area	D	H	R	τ_g	τ_t	Psij
1 – 2	1	0.6	5	0.05	0.2	0.5	1.918
2 – 3	2	0.9	4	0.0625	0.3	0.6	1.720
3 – 1	3	0.8	6	0.08	0.25	0.4	1.861

The three units operate in parallel at the nominal frequency of 60Hz. A load change of 200MW (0.2pu) occurs at area 2. Construct a SIMULINK block diagram to obtain the response of the frequency deviation, mechanical power deviation, and tie line power deviation responses for the two types of control: the primary LFC control, and the tie line bias control (ACE).

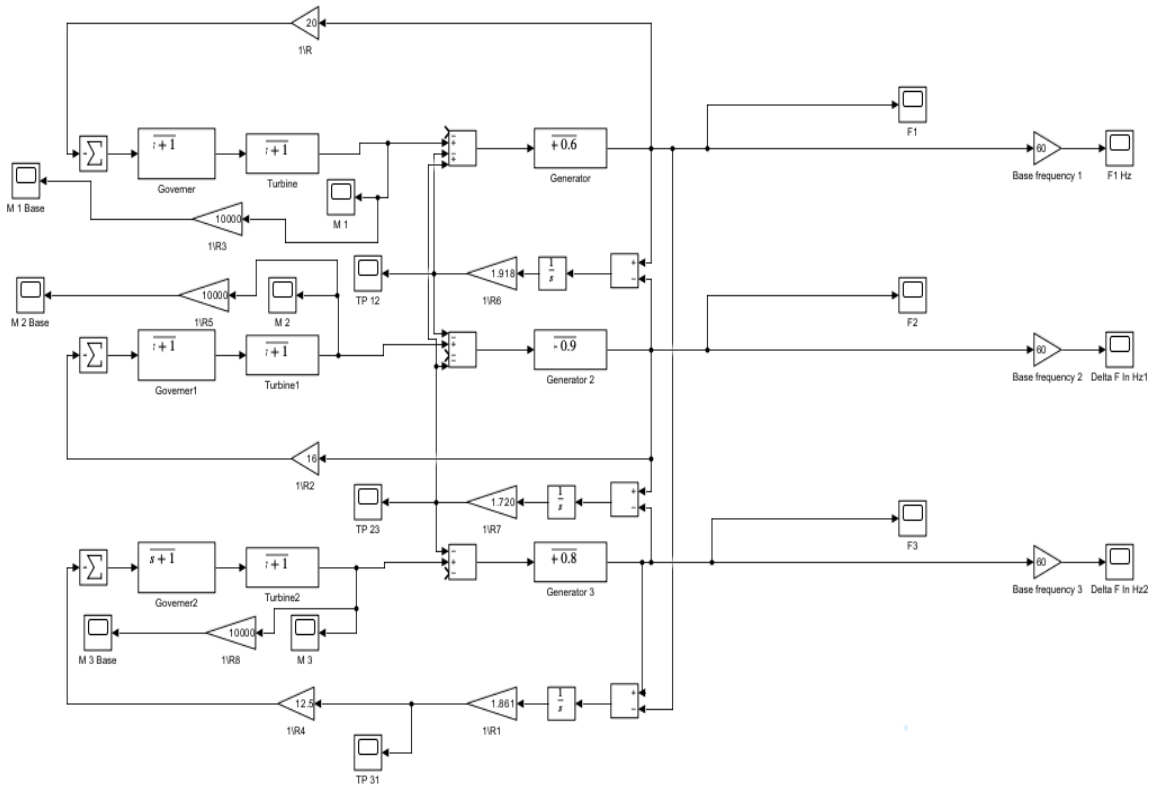


Figure 4.1: Three area load frequency control 9LFC Before adding controller

And after applying the equations for the transfer function of the block diagram for the inertia and load for each area we get:

$$\frac{1}{2H_1 s} + D_1 = \frac{1}{10s + 0.6}$$

$$\frac{1}{2H_2 s} + D_2 = \frac{1}{8s + 0.9}$$

$$\frac{1}{2H_3 s} + D_3 = \frac{1}{12s + 0.8}$$

And to calculate the speed regulation for each area in the simulation:

And for the turbine transfer function

The per unit change in area 2 is given by:

The per unit steady state frequency deviation is:

$R3$Steady state frequency deviation in Hz:

$\Delta factual$ = Δfpu x f base = −0.2358Hz

$factual$ = f original + Δf actual = 59.764Hz

The Mechanical power generated:

$\Delta Pm1 = (\frac{-\Delta\omega}{R1}) = 0.0786pu$

$\Delta Pm1 = 0.0786x1000 = 78.6MW$

$\Delta Pm2 = (\frac{-\Delta\omega}{R2}) = 0.0628pu$

$\Delta Pm2 = 0.0628x1000 = 62.8MW$

$\Delta Pm3 = (\frac{-\Delta\omega}{R3}) = 0.0491pu$

$\Delta Pm1 = 0.0491x1000 = 49.1MW$

These values must be compared with the results obtained from the simulation, and they should closely match

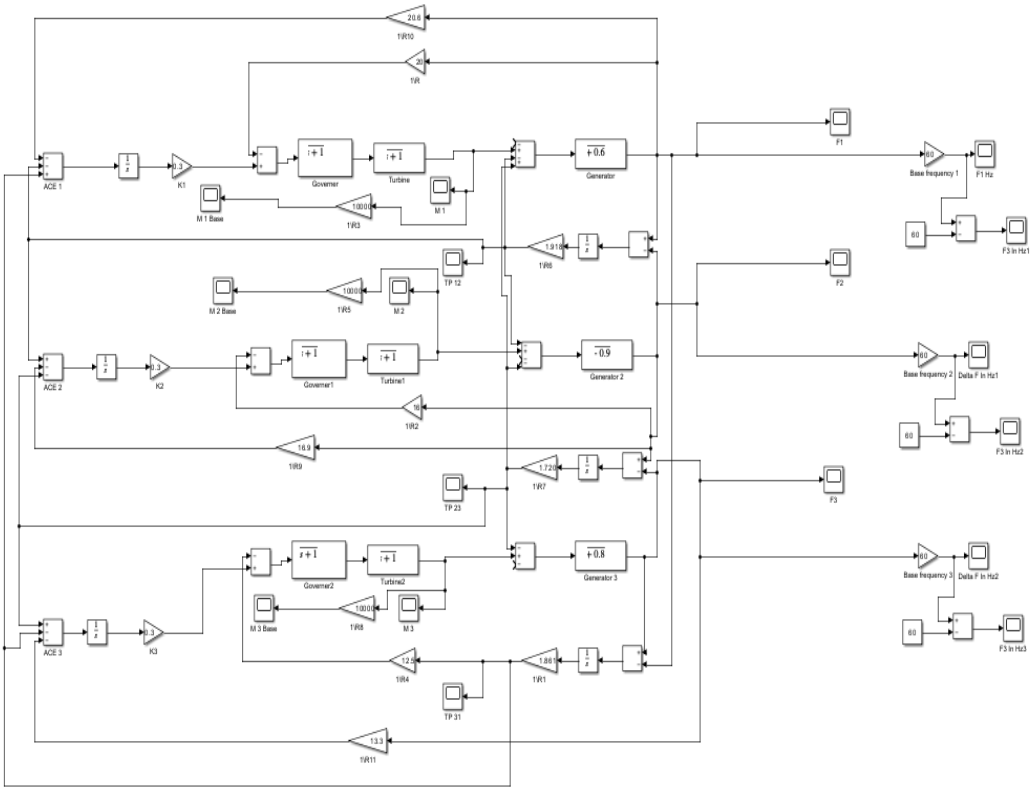


Figure 4.2: three area load frequency control (LFC) after adding controller.

4.2 Comparing the result before and after implementing frequency control:

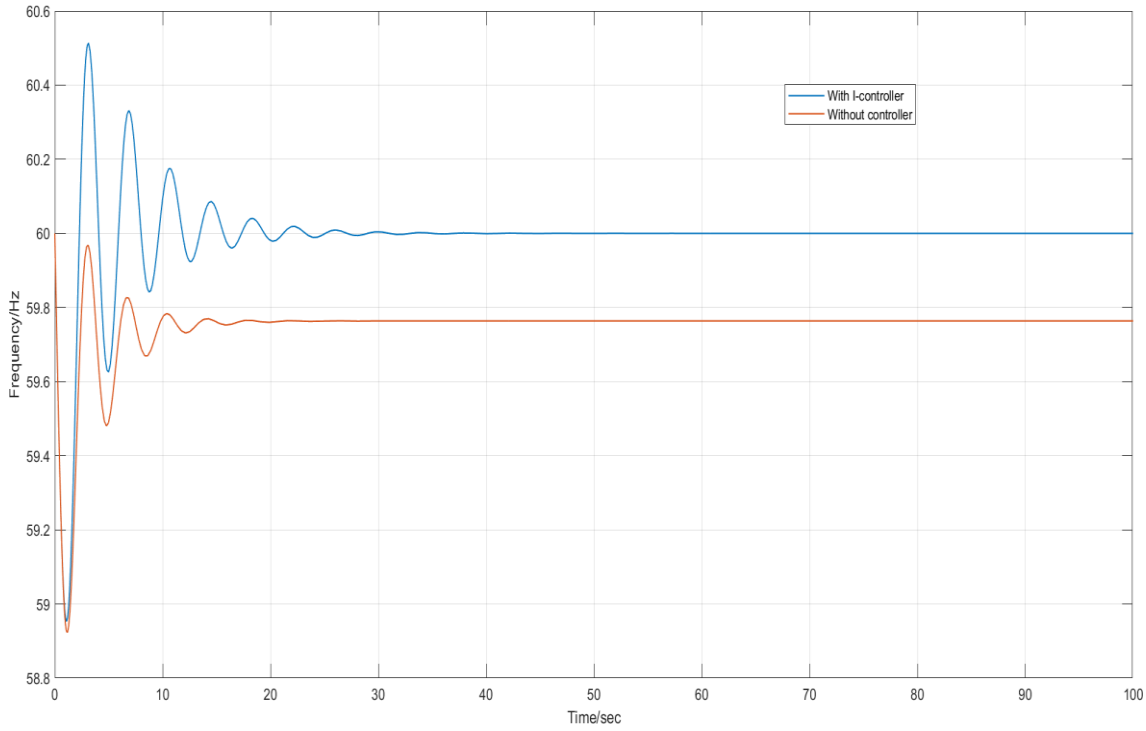


Figure 4.3: comparing between frequency in Hz.

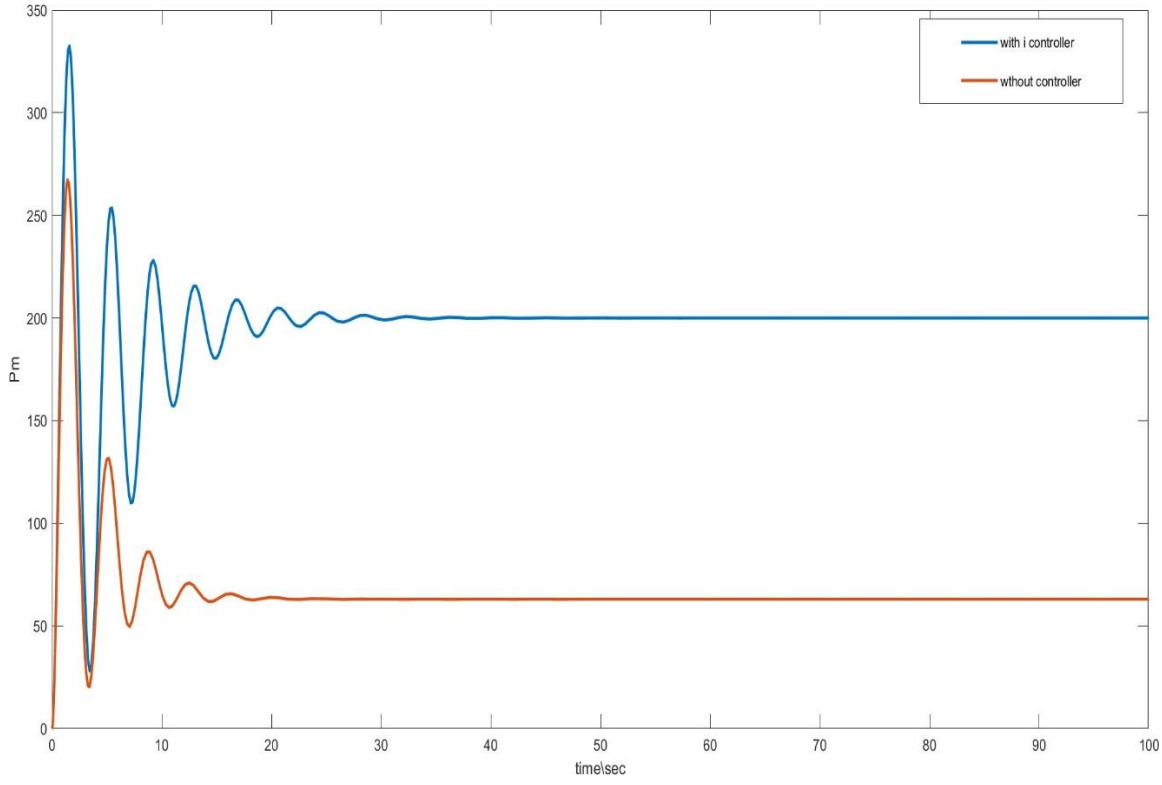


Figure 4.4: Comparing between mechanical power.

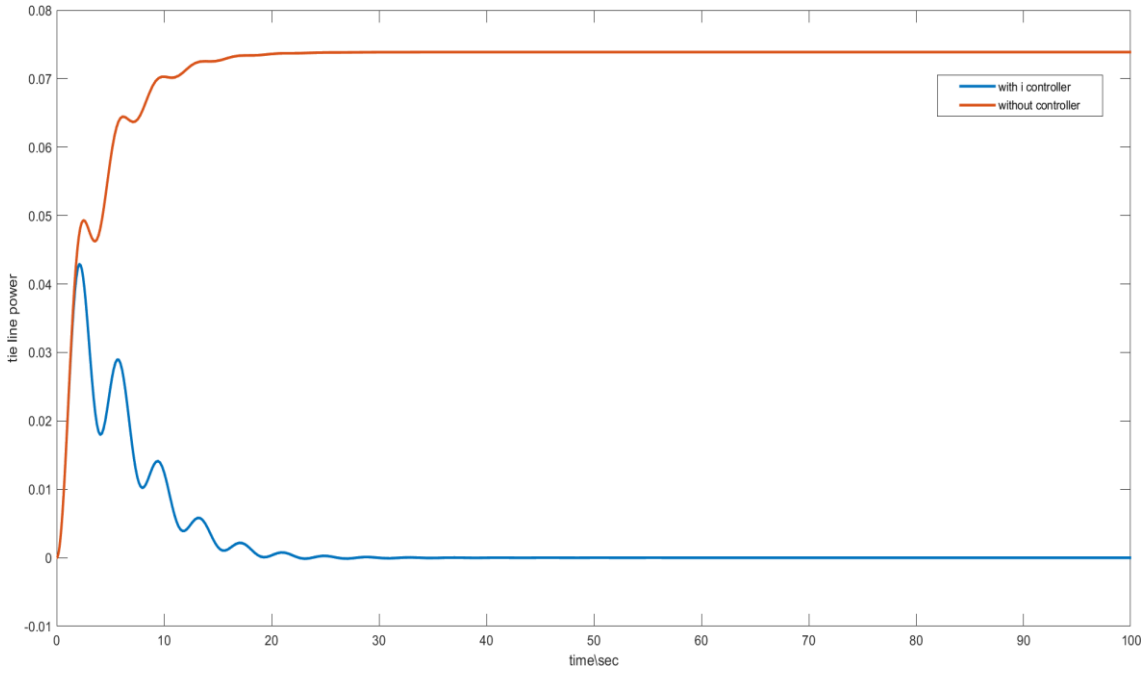


Figure 4.5:Comparing between tie line power

$$B_1 = \frac{1}{R_1} + D_1 = 20.6$$

$$B_2 = \frac{1}{R_2} + D_2 = 16.9$$

$$B_3 = \frac{1}{R_3} + D_3 = 13.3$$

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